

Mathematical Modelling — Tutorial 1

How these tutorials work. Each tutorial has pen-and-paper parts and a *computational task* that you carry out in whatever language you prefer (MATLAB is recommended; Python or Julia are equally fine). The point of the code is not the language, but it is that a classmate can regenerate your figure from your folder. Each week you add a folder `weekNN/<username>/` to the course tutorials repository on GitHub, containing a half-page `report.md`, the figure(s) requested, the code that produced them, and a short `README` giving the command to regenerate them. You submit this as a pull request and review one classmate's in return (see the repository's `CONTRIBUTING.md`). Parts marked *Challenge* are optional, for the motivated or when time allows. Chapter and appendix references are to the course notes. Every `report.md` should also close with one sentence of *interpretation*: what does your result mean for the real system you modelled, and not merely as a piece of mathematics?

Before you start — GitHub setup. The computational work is submitted through GitHub. *This week, before the tutorial:* (1) create a free GitHub account if you do not already have one, at github.com/signup; (2) email me your GitHub username so I can add you to the course organisation — it is *private*, so you must be invited and cannot join on your own; (3) accept the invitation that then arrives by email. Full step-by-step instructions — creating the account, joining the organisation, installing Git, and a first-time guide for those new to it — are in the “**Start here**” document in the **Modules** section on Canvas. *Read it first if you have never used GitHub.* You do not need to arrive knowing Git: in the tutorial we will run a short, hands-on introduction to the few commands you need (clone, commit, push, and opening a pull request), and by the end you will have made your first submission. All you need beforehand is an account and an accepted invitation.

Tutorial 1 (Week 1 — Dimensional analysis in practice)

Notes: Chapter on dimensional analysis. This week also serves as your onboarding to the course repository.

1. Rederive, by dimensional analysis alone, the period of a simple pendulum $T = \sqrt{L/g} \Phi(\theta_0)$, stating clearly what the method fixes and what it leaves in the unknown function Φ .
2. Pick one further system — the frequency of a mass on a spring, the speed of deep-water waves, or the draining time of a tank — and carry out the same analysis: list the variables, form the dimensionless groups, and state precisely what is predicted and what is left undetermined.
3. A rigid-body (compound) pendulum — an extended rigid object swinging about a pivot — has small-amplitude period $T = 2\pi\sqrt{L_{\text{eff}}/g}$, where g is gravity and $L_{\text{eff}} = I/(mR)$ is its *effective length*, built from the body's mass m , its moment of inertia I about the pivot, and the pivot-to-centre-of-mass distance R (see the notes, Chapter 6). Without computing any moment of inertia, explain why dimensional analysis already guarantees the $\sqrt{L_{\text{eff}}/g}$ dependence.

Computational task. Taylor's blast-wave argument predicts $R = C(Et^2/\rho)^{1/5}$ for the radius R of a blast at time t . Generate a synthetic “photographic” data set: fix $\rho = 1.2$ and take the constant $C = 1$ (its theoretical value), choose an energy E , compute $R(t)$ at a dozen times, and

add a few percent of random noise to mimic measurement error. Now *pretend you do not know* E : fit a straight line to $\log R$ versus $\log t$, check that the slope is close to $2/5$, and recover E from the intercept (with $C = 1$ and ρ known, the intercept gives E/ρ). Report how close your estimate is to the true E .

Repo deliverable.

1. Your fitting code;
2. a log-log plot showing the data and the fitted line;
3. a `README` giving the command to regenerate the figures, and;
4. a `report.md` stating the recovered slope and energy — closing with what this shows: that a handful of photographs, plus dimensional analysis alone, suffice to weigh an explosion.

This is your first pull request — get the workflow working now, while the mathematics is easy.